

SOFTWARE ENGINEERING & TESTING ASSIGNMENT

SDLC, V-Model, and RAD Model Analysis

Question 1

Draw the V-Model diagram on paper or using any digital tool, and clearly label each phase with both the developer and tester roles involved at each step.

Ans: V-Model Architecture Diagram



Explanation of Each Phase

1. Requirements Analysis ↔ Acceptance Testing

- **Developer Role:** Understands customer requirements and creates requirement documents.
- **Tester Role:** Prepares Acceptance Test Cases and checks whether the final software meets customer requirements.

2. System Design ↔ System Testing

- **Developer Role:** Designs the complete system architecture.
- **Tester Role:** Tests the entire application as a whole and verifies all features are working correctly.

3. High-Level Design (HLD) ↔ Integration Testing

- **Developer Role:** Designs how different modules interact.
- **Tester Role:** Tests communication between modules and ensures modules work together properly.

4. Low-Level Design (LLD) ↔ Unit Testing

- **Developer Role:** Designs individual functions and classes.
- **Tester/Developer Role:** Tests each unit or function separately to ensure each component works correctly.

5. Coding

- **Developer Role:** Writes the actual program code.
- **Tester Role:** No testing is performed at this stage. Testing starts after coding is completed.

Question 2

Pick any app you use daily (like Zomato, Instagram, or Paytm) and write a short table mapping each SDLC phase (Requirement, Design, Coding, Testing, Deployment, Maintenance) to a real activity that would happen for that app.

Ans: SDLC Phases for Zomato App

SDLC Phase	Real Activity in Zomato App
Requirement	The company decides that users should be able to track their food orders in real time.
Design	UI/UX designers create screens showing the delivery partner's location and order status.
Coding	Developers write code for order tracking, maps integration, and status updates.
Testing	Testers check whether the tracking feature works correctly and updates the delivery location properly.
Deployment	The new order tracking feature is released to users through the Zomato app update.
Maintenance	Developers fix bugs, improve tracking accuracy, and update the feature based on user feedback.

Question 3

For each phase of the V-Model, list one specific testing activity that should be performed, using examples from a food delivery app (like Swiggy or Zomato).

Ans: V-Model Testing Activities Mapping

V-Model Phase	Testing Activity	Example in Food Delivery App
Requirements Analysis	Review Requirements	Check that requirements clearly state users can place orders, track orders, and make payments.
System Design	Create System Test Cases	Prepare test cases for complete order flow from restaurant selection to food delivery.
High-Level Design (HLD)	Create Integration Test Cases	Verify that payment, order, and notification modules work together correctly.
Low-Level Design (LLD)	Create Unit Test Cases	Test individual functions such as discount calculation and delivery charge calculation.
Coding	Unit Testing	Developers test each code module separately, such as login and payment functions.
Unit Testing Phase	Verify Individual Components	Check if the promo code feature calculates the correct discount.
Integration Testing Phase	Test Module Interaction	Verify that after payment is successful, the order is automatically placed.
System Testing Phase	End-to-End Testing	Test complete order placement, payment, tracking, and delivery process.
Acceptance Testing Phase	Validate User Requirements	Confirm that users can order food and track deliveries as specified in requirements.

Question 4

Compare the V-Model and RAD Model by listing 3 strengths and 2 weaknesses of each, focusing on how testing is handled in both models.

Ans: Model Comparison Matrix

Model	Strengths	Weaknesses
V-Model	1. Testing is planned from the beginning of the project. 2. Defects are found early, reducing fixing costs. 3. Every development phase has a corresponding testing phase, improving quality.	1. Changes in requirements are difficult to handle. 2. Not suitable for large projects where requirements change frequently.
RAD Model	1. Faster development and testing cycles. 2. Users provide continuous feedback during development. 3. Quick identification and correction of issues through prototypes.	1. Testing may be rushed because of tight deadlines. 2. Documentation and detailed testing can be less thorough.